Prodigy Anesthesia Quick Review

Acromegaly

What is acromegaly and what is the most common cause?	Acromegaly is the excessive release of growth hormone. It is most commonly due to an adenoma in the anterior pituitary gland. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 404.*
What are the most common clinical features of acromegaly?	The increased growth hormone levels stimulate the release of insulin-like growth factor I. These two hormones result in an increase in the proliferation of cartilage, bone, protein synthesis, and lipolysis. It also decreases insulin sensitivity and promotes the retention of sodium. As a result of these effects, common manifestations include enlargement of the jaw, hands, and feet as well as hypertension and diabetes mellitus. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 22.*
How does acromegaly affect the integumentary system?	Acromegaly results in thick, oily skin and hyperhidrosis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 404.*
How does acromegaly affect the respiratory system?	Lung volumes are typically increased and V/Q mismatching may occur. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 404.*
How is the musculoskeletal system affected by acromegaly?	Osteoarthritis and osteoporosis are common. Skeletal muscle weakness is a typical finding and may manifest as an increased sensitivity to muscle relaxants. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 404.*
How does acromegaly affect the peripheral nervous system?	Overgrowth of soft tissue results in trapping of peripheral nerves. Carpal tunnel syndrome is a common finding, as is decreased ulnar artery flow due to compression by soft tissue. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 404.*
What airway abnormalities are associated with acromegaly?	Acromegaly results in numerous anatomical changes that can complicate airway management including prolonged mandible, overgrowth of pharyngeal tissue, recurrent laryngeal nerve damage, macroglossia, and decreased subglottic diameter. Hoarseness or stridor are indications that the upper airway is probably involved. Tracheal compression and deviation can occur due to thyroid enlargement. Sleep apnea is often present. The risk of respiratory failure is three times higher in patients with acromegaly. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 22.*

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What are the cardiac manifestations of acromegaly?	The chronic hypertension results in ventricular dysfunction, enlarged heart, congestive heart failure, ischemic heart disease, and dysrhythmias. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 22.*

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